

Lec 8: Reps of SL_2 & $\mathfrak{sl}_2$, IV.

1) Representations of $\mathfrak{sl}_2(F)$, finished

2) Representations of $SL_2(F)$

1.0) Recap & roadmap

Let F be an algebraically closed field of $\text{char} = p > 2$.

We saw in Sec 1.1 of Lec 7 that the elements $e^p, h^p - h, f^p \in \mathcal{U}(\mathfrak{sl}_2)$ are central. We also classified (Sec 2.2 of Lec 7) the $\mathfrak{g}$ -irreps where these elements act by p -character $\lambda = (\lambda_e, \lambda_h, \lambda_f)$ as follows

(1) $(0,0,0); (0,a,0), a \neq 0; (0,0,1)$

Let's explain why the case of general λ reduces to one of these.

Let $\mathfrak{g}$ be a Lie algebra, $\varphi: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ a representation & $\tau: \mathfrak{g} \xrightarrow{\sim} \mathfrak{g}$ a Lie algebra automorphism. Then $\varphi^\tau := \varphi \circ \tau$ is also a representation, irreducible iff φ is.

An important example of φ is as follows. Let G be an algebraic group w. $\text{Lie}(G) = \mathfrak{g}$ & $g \in G$. We have the automorphism $a_g: G \rightarrow G$ $g' \mapsto gg'g^{-1}$ & its tangent map $\text{Ad}(g): \mathfrak{g} \rightarrow \mathfrak{g}$, now a Lie algebra automorphism. If $G \subset GL(V)$, then $\text{Ad}(g)$ is the matrix conjugation by g . See Example 2 in Sec 1.2 of Lec 4. We write φ^g for $\varphi^{\text{Ad}(g)}$ & V^g for the corresponding $\mathcal{U}(\mathfrak{g})$ -module. We'll see that

(*) $\forall \mathfrak{sl}_2$ -irrep V we can find $g \in SL_2$ (by means of Linear algebra) s.t. the p -character of V^g is from (1).

This will complete our classification of irreps.

1.1) Move on elements $x^p - x^{[p]}$

In this section $\mathfrak{g} = \text{Lie}(G)$, where G is an algebraic group.

Recall that we have the map $\mathfrak{g} \xrightarrow{\iota} Z(\mathcal{U}(\mathfrak{g})), x \mapsto x^p - x^{[p]}$, Sec 1.1 of Lec 7. Let's investigate its compatibilities with the G -representation structure on $\mathfrak{g}$. Note that since G acts on $\mathfrak{g}$ by automorphisms (via Ad), we get a G -representation on $\mathcal{U}(\mathfrak{g})$ by algebra automorphisms (also called adjoint). Every automorphism preserves $Z(\mathcal{U}(\mathfrak{g}))$.

Thm: a) $\iota(ax) = a^p \iota(x), \forall a \in \mathbb{F}, x \in \mathfrak{g}$

$$1) \iota(\text{Ad}(g)x) = \text{Ad}(g)\iota(x) \quad \forall g \in G, x \in \mathfrak{g}$$

$$2) \iota(x+y) = \iota(x) + \iota(y) \quad \forall x, y \in \mathfrak{g}.$$

Proof:

a) $\Leftarrow$ both $x^p, x^{[p]}$ are p th powers in suitable associative algebras

1): Since $\text{Ad}(g)$ is an automorphism of $\mathcal{U}(\mathfrak{g})$, $(\text{Ad}(g)x)^p = \text{Ad}(g)(x^p)$.

Now assume $G \subset GL(V)$. Then $\text{Ad}(g) = \text{conjugation w. } g$ so $(\text{Ad}(g)x)^{[p]} = (gxg^{-1})^p = gx^pg^{-1} = \text{Ad}(g)(x^{[p]})$ (equalities in $\text{End}(V)$). 1) follows.

2): will be proved later □

1.2) p -characters, conceptually.

We now assume that $G = SL_2$.

Let V be a (fin. dim.) $\mathfrak{g}$ -irrep $\leadsto$ central character $\chi_V: Z(\mathcal{U}(\mathfrak{g}))$

$\rightarrow \mathbb{F} \leadsto p$ -character $\chi_V \circ \iota: \mathfrak{g} \rightarrow \mathbb{F}$, so that $\chi_x = \chi_V \circ \iota(x), x \in \{e, h, f\}$.

2)

Exercise: $X_{Vg} = X_V \circ \text{Ad}(g)$ (hint: from constructions).

Recall the automorphism $\text{Fr}: \mathbb{F} \rightarrow \mathbb{F}, x \mapsto x^p$.

Definition: We say that a function $X: \mathfrak{g} \rightarrow \mathbb{F}$ is **Fr-linear** if $X(x+y) = X(x) + X(y)$ & $X(ax) = \text{Fr}(a)X(x) \ \forall x, y \in \mathfrak{g}, a \in \mathbb{F}$.

The set of such functions is denoted by $\mathfrak{g}^{*(1)}$.

Example: Thx to 0) & 2) of Theorem, $X_V \circ \iota \in \mathfrak{g}^{*(1)}$.

Some basic remarks on $\mathfrak{g}^{*(1)}$ are in order.

i) Just as with linear functions, an element of $\mathfrak{g}^{*(1)}$ is determined by its values on a basis, say e, h, f .

ii) G acts on $\mathfrak{g}^{*(1)}$: $g.X := X \circ \text{Ad}(g^{-1})$. We care about this b/c $g^{-1} \cdot (X_V \circ \iota) = X_V \circ \iota \circ \text{Ad}(g) = [1 \text{ of Thm}] = X_V \circ \text{Ad}(g) \circ \iota = [\text{Exercise}] = X_{Vg} \circ \iota$

iii) we have a G -equivariant bijection $\mathfrak{g}^* \xrightarrow{\sim} \mathfrak{g}^{*(1)}: d \mapsto \text{Fr} \circ d$.

On the other hand, we claim that $\exists$ G -representation isomorphism $\mathfrak{g} \xrightarrow{\sim} \mathfrak{g}^*$. Indeed, we have a symmetric bilinear form $(; \cdot): \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{F}$:

$(x, y) = \text{tr}(xy)$. It's non-degenerate - its matrix in the basis e, h, f is $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ & G -invariant: $(\text{Ad}(g)x, \text{Ad}(g)y) = \text{tr}(gxg^{-1}gyg^{-1}) = (x, y)$

This yields an isomorphism $\mathfrak{g} \rightarrow \mathfrak{g}^*: x \mapsto [y \mapsto \text{tr}(xy)]$ & hence

$\mathfrak{g} \rightarrow \mathfrak{g}^{*(1)}: x \mapsto [y \mapsto \text{Fr}(\text{tr}(xy))]$, both are G -equivariant.

Example: Consider the element $\frac{1}{2} \text{Fr}^{-1}(a)h \in \mathfrak{g}$ for $a \in \mathbb{F}$. Its image in $\mathfrak{g}^{*(1)}$ vanishes on e, f and sends h to $\text{Fr}(\frac{1}{2} \text{Fr}^{-1}(a) \text{tr}(h^2)) = a$. We get two of p -characters in (1) of Sec 1.0. The third, $(0, 0, 1)$, is the image of e (exercise).

Now we can establish (*). By the JNF thm, every G -orbit in $\mathfrak{g}$ contains either bh ($b \in \mathbb{F}$) or e . Since $a \mapsto \frac{1}{2} \text{Fr}^{-1}(a): \mathbb{F} \rightarrow \mathbb{F}$ is bijective, and $\mathfrak{g} \rightarrow \mathfrak{g}^{*(1)}$ is G -equivariant, we see that the G -orbit of any element of $\mathfrak{g}^{*(1)}$ contains one of elements in (1). ii) implies (*).

2) Representations of $SL_2(\mathbb{F})$

We now proceed to studying the finite dimensional rational representations of $SL_2(\mathbb{F})$ (for simplicity, $\text{char } \mathbb{F} \neq 2$). We'll see that

- if $\text{char}(\mathbb{F}) = 0$, then the rep. theories of SL_2 & $\mathbb{A}^1 SL_2$ are the same.
- while for $\text{char } \mathbb{F} > 0$, they are quite different: in fact, oversimplifying one can say that the representations of $SL_2(\mathbb{F})$ are closer to the characteristic 0 story (but are not completely reducible).

2.1) Some (ir)reducible representations

Consider the representation of $G = SL_2$ in homogeneous degree n polynomials, $M(n) := \text{Span}_{\mathbb{F}}(x^n, x^{n-1}y, \dots, y^n)$,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot f(x, y) = f(ax+cy, bx+dy).$$

Lemma: $M(n)$ is rational & its tangent representation is given

by $e(x^i y^{n-i}) = (n-i)x^{i+1}y^{n-i-1}$, $f(x^i y^{n-i}) = ix^{i-1}y^{n-i+1}$, $h(x^i y^{n-i}) = (2i-n)x^i y^{n-i}$.
 where $e, h, f \in \mathcal{SL}_2$ is a basis as in Example in Sec 3.1.

Proof:

For $n=1$ we get the tautological representation of $\mathcal{SL}_2 / \mathcal{SL}_2$.

In general, $M(n) = \text{Sym}^n M(1)$ hence a quotient of $M(1)^{\otimes n}$, hence rational. Under the epimorphism $M(1)^{\otimes n} \twoheadrightarrow M(n)$ we have $x^{\otimes i} \otimes y^{\otimes n-i} \mapsto x^i y^{n-i}$, hence $e(x^{\otimes i} \otimes y^{\otimes n-i}) = \sum_{j=1}^{n-i} x^{\otimes i} \otimes y^{\otimes j-1} \otimes x \otimes y^{\otimes n-i-j} \mapsto (n-i)x^{i+1}y^{n-i-1}$.
 The formulas for h & f are obtained similarly $\square$

Proposition: $M(n)$ is irreducible and isomorphic to $\mathcal{L}(n)$ over $\mathbb{C}$ if

- 1) $\text{char } \mathbb{F} = 0$
- 2) or $\text{char } \mathbb{F} > 2$ & $n < p$.

Proof:

The map $\mathcal{L}(n) \rightarrow M(n)$, $f^i v_\lambda \mapsto \frac{1}{i!} x^{n-i} y^i$ ($i=0, \dots, n$) is an isomorphism of $\mathcal{SL}_2$ -reps (**exercise**: use Lemma) so $M(n)$ is irreducible as a $\mathfrak{g}$ -representation. By Section 2 of Lec 5, $M(n)$ is irreducible over $\mathbb{C}$ $\square$

Example: $M(p)$ is not irreducible if $\text{char } \mathbb{F} = p$. Indeed, $\text{Span}_{\mathbb{F}}(x^p, y^p) \subset M(p)$ is a subrepresentation, e.g. $\begin{pmatrix} a & b \\ c & d \end{pmatrix} x^p = (ax+cy)^p = a^p x^p + c^p y^p$. More generally, if $n \geq p$, then $M(n)$ is irreducible $\Leftrightarrow n = p^k - 1$ for some $k \geq 1$ (shouldn't be clear at this point).

2.2) Classification in characteristic 0

Theorem: Assume $\text{char } \mathbb{F} = 0$. Then every finite dimensional rational

G -representation, V , is completely reducible and the irreducibles are classified by $\mathbb{Z}$ via $n \mapsto M(n)$.

Proof: As a $\mathfrak{g}$ -representation, $V \simeq \bigoplus_{i=1}^k M(n_i)$ for some n_i thx to the results of Lecture 6 & Proposition in Sec 2.1. Now we invoke Answer 3 in Sec 2 of Lec 5 to deduce an isomorphism of G -reps $\square$

2.3) Irreducibles in characteristic p .

Assume now $\text{char } \mathbb{F} = p > 2$. Recall that to a representation $\varphi: G \rightarrow GL(V)$ we can assign its Frobenius twist $\varphi^{(1)} := \text{Fr} \circ \varphi$. Note that $\text{Fr}: GL(V) \rightarrow GL(V)$ is defined using an isomorphism $GL(V) \rightarrow GL_n$ that we get identifying V w. $\mathbb{F}^n$ but different identifications (i.e. choices of basis) result in conjugating Fr by an element of $GL(V)$ hence in isomorphic representations (exercise).

Exercise: Define $\text{Fr}: \mathbb{F}^n \rightarrow \mathbb{F}^n, (a_1, \dots, a_n) \mapsto (a_1^p, \dots, a_n^p)$. Prove that $U \subset \mathbb{F}^n$ is a subrep'n for $\varphi \iff \text{Fr}(U)$ is a subrep'n for $\varphi^{(1)}$. In particular, φ is irreducible $\iff \varphi^{(1)}$ is.

Example: $\text{Span}_{\mathbb{F}}(x^p, y^p) \simeq M(1)^{(1)}$ as G -representations.

Proposition: If V is irreducible, then $M(i) \otimes V^{(1)}$ is irreducible $\forall i \in \{0, \dots, p-1\}$.

Proof: Note that $\mathfrak{g}$ acts on $V^{(1)}$ by 0 (by Answer 2 in Sec 2 of Lec 5). So $M(i) \otimes V^{(1)} (\simeq M(i)^{\oplus \dim V})$ is a completely reducible $U(\mathfrak{g})$ -module.

Prop 2.17 in Chapter 0 of notes on the website allows to describe its g -submodules: they are of the form $M(i) \otimes V'$ for a subspace $V' \subset V^{(1)}$. Note that $g(M(i) \otimes V') = M(i) \otimes gV'$. So $M(i) \otimes V'$ is G -stable $\Leftrightarrow V' \subset V^{(1)}$ is G -stable. Since V (hence $V^{(1)}$) is irreducible, we are done. $\square$

This gives rise to the following inductive construction. For $k \geq 0$, we write $\cdot^{(k)}$ for $\cdot^{(1)}$ repeated k times.

Exercise: For rational representations V_1, V_2 of G , we have $(V_1 \otimes V_2)^{(1)} \simeq V_1^{(1)} \otimes V_2^{(1)}$ (hint: look at matrix coefficients)

Corollary (Steinberg tensor product theorem) For $0 \leq \lambda_0, \dots, \lambda_k \leq p-1$, the G -representation $M(\lambda_0) \otimes M(\lambda_1)^{(1)} \otimes \dots \otimes M(\lambda_k)^{(k)}$ is irreducible.

In the next lecture we'll see that every irreducible rational representation of G is isomorphic to exactly one of these tensor products.